PEDIATRIC CHRONIC PAIN

Pediatric Pain PRN Curriculum

The MAYDAY Fund

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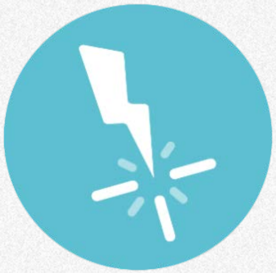
Objectives

- Describe the prevalence and costs of chronic pain experienced by children and adolescents.
- Explain the difference between an acute and chronic pain management plan, including the unique challenges of managing chronic pain.
- Outline at least 5 realistic treatment goals and different corresponding treatments to help achieve these goals.

What is
Chronic Pain?

Chronic Pain



Chronic pain is typically defined by:

- time (> 3 months, IASP)
- persistence (beyond the time expected for healing)
- consequence of ongoing or progressive disease (for example, spasticity, sickle cell disease, arthritis)
- impaired function without apparent biological value

Chronic pain may also be episodic with recurrent bouts of pain at least every 3 months.

Uncontrolled acute pain has been recognized as a significant risk factor for the development of chronic pain.

Disease-related pain:

- Sickle cell disease, Epidermolysis Bullosa, Osteogenesis Imperfecta, Rheumatological conditions, Cancer and treatment related pain (e.g. chemotherapy).

Injury-related pain:

- Burns, Fractures, Post-surgery (e.g. phantom limb pain, scar tissue, nerve damage), Complex regional pain syndrome (e.g. post-fracture or sprain).

Non-specific (Unexplained/Chronic Benign Pain):

- Headache, Recurrent abdominal pain, Pain Amplification syndrome, Complex regional pain syndrome, Low back pain, Widespread chronic pain, Chronic fatigue syndrome, Fibromyalgia.

Adjustment Disorders:

- Somatoform disorder, Pain Disorder, Conversion Disorder (very rare).

Pediatric Chronic Pain

Cost of pediatric chronic pain in the U.S. has been calculated at >$19.5 billion/year

(Groenewald, et al., 2014)

Prevalence and significance

- Estimates suggest that 20-46% of US children and adolescents suffer from chronic pain; but
- Reported prevalence rates vary from 4–88%, with the highest pain prevalence rates for
  - headaches (up to 83%),
  - abdominal pain (up to 53%), and
  - musculoskeletal pain (up to 36%)
- More common in females
- Most common in early teen years

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Types of Chronic Pain

- Headaches
- Abdominal
- Musculoskeletal
- Back

Headaches

Most common persistent pain complaint in pediatrics. (8-83%)

Occurs in all ages of children

- 4% of preschoolers
- 10% of school-aged children (girls 1:1 boys)
- >16% of adolescents (more girls 3:1 boys)

Types

- Tension
- Migraine
- Pseudo Tumor Cerebri
- Persistent >15 days/month for >3 months

Almost every child reports having had a headache in the last 3 months. Most children with recurrent headaches have a family history of headaches. Headache prevalence also increases with age. Prevalence for boys and girls is similar before 12 years of age; but after 12 years, prevalence for girls is triple that of boys.

The etiology of the headache should dictate the treatment plan. Therefore, the goal of treating migraines and other headaches without known cause is pain relief or reduction to minimize associated functional disability.

Abdominal

The numbers

- Affects 4-53% of children
- Accounts for 25% of pediatric gastroenterology office visits and may be a precursor to IBS
- Average cost of $6,000 for diagnostic work up for functional dyspepsia per child

Functional abdominal pain (Rome III criteria)

- Episodic or continuous abdominal pain >3 months
- Typically described as nonradiating pain around umbilicus that lasts 1-3 hours
- Accompanying symptoms may include pallor, diaphoresis, nausea, vomiting, sleep disturbances, and changes in oral intake
- Insufficient criteria for other functional gastrointestinal disorders
- No evidence of an inflammatory, anatomic , metabolic, or neoplastic process that explains the patient's symptoms
- Other terms that have erroneously been used interchangeably are nonorganic abdominal pain, psychogenic abdominal pain, and chronic recurrent abdominal pain (CRAP).

At least 25% of patients have some loss of daily function and other pain, such as headache or limb pain.

Abdominal

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Rome Criteria

The Rome criteria for a diagnosis of functional abdominal pain includes episodic or continuous abdominal pain for greater than 3 months without identified pathology on physical examination and a negative stool for occult blood.

Functional abdominal pain is described as nonradiating pain that lasts 1 to 3 hours and is typically located around the umbilicus.

Accompanying symptoms may include altered bowel pattern, pallor, diaphoresis, nausea, vomiting, sleep disturbances, and changes in oral intake. In at least 25% of cases, patients also have some loss of daily function and other somatic symptoms, such as headache or limb pain.

Other terms that have erroneously been used interchangeably with functional abdominal pain are nonorganic abdominal pain, psychogenic abdominal pain, and chronic recurrent abdominal pain.

Other common chronic abdominal pain conditions include: functional dyspepsia, irritable bowel syndrome (IBS), and abdominal migraine.

Musculoskeletal

Chronic musculoskeletal pain can be categorized into three subtypes:

- Injury-related,
- Illness-related, or
- Primary musculoskeletal pain.

Pain can be localized to specific joints or muscle groups, like ankle injuries or back pain, or in multiple joints or muscle groups, like arthritis.

Connective tissue disorders, such as Ehlers Danlos or other hypermobility disorders may be associated with generalized musculoskeletal pain.

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Musculoskeletal pain may also be diffuse as with fibromyalgia. Sometimes musculoskeletal pain appears as both diffuse and idiopathic in nature, making the pain difficult to characterize. Diffuse types of musculoskeletal pain have a high comorbidity with depressive symptoms (Kashikar-Zuck & Ting, 2014). Researchers believe the stress-adaptation response is altered in children with fibromyalgia, possibly due to immune mediated processes (Kashikar-Zuck & Ting, 2014).

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Musculoskeletal

Non-inflammatory Musculoskeletal Pain Syndromes

- Growing Pains
- Patello-Femoral Syndrome
- Benign Joint Hypermobility Syndrome

Neuropathic Pain Syndromes

- Complex Regional Pain Syndrome:
  - Ongoing burning of limb after an injury or immobilization,
  - Leading to allodynia, hyperalgesia and autonomic nervous system dysfunction
  - Important to diagnose and start treatment early
    - Multi-disciplinary approach
    - Desensitization, gradual increase weight bearing, ROM
    - May need analgesics to tolerate therapy
- Juvenile Primary Fibromyalgia Syndrome (JPFS)

Neuropathic pain is thought to be less common in children, but there are no current epidemiological studies (Howard, Weiner & Walker, 2014).

Children experience neuropathic pain from nerve injury as a result of surgery or trauma (e.g., phantom limb pain or complex regional pain syndrome (CRPS), autoimmune or degenerative neuropathies (e.g., Guillain-Barré syndrome or Charcot-Marie-Tooth disease), and neuropathies related to cancer or cancer treatment. CRPS manifests as severe localized pain with allodynia, swelling, and color and temperature changes. CRPS typically affects a single extremity. A minor to severe injury, trauma, infection, or stressful event precipitates the syndrome. A precipitating event to any extremity can also cause CRPS to recur (Howard, Weiner & Walker, 2014).

Musculoskeletal

Inflammatory musculoskeletal pain

Juvenile arthritis (JA) effects nearly 300,000 US children

- oligoarthritis,
- polyarthritis,
- systemic, enthesitis-related,
- juvenile psoriatic arthritis or
- Undifferentiated.

Juvenile arthritis (JA) is not a disease in itself. JA is an umbrella term used to describe the many autoimmune and inflammatory conditions or pediatric rheumatic diseases that can develop in children ages 16 and younger. Although the various types of juvenile arthritis share many common symptoms, like pain, joint swelling, redness and warmth, each type of JA is distinct and has its own special concerns and symptoms. Some types of juvenile arthritis affect the musculoskeletal system, but joint symptoms may be minor or nonexistent. Juvenile arthritis can also involve the eyes, skin, muscles and gastrointestinal tract.

Musculoskeletal

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Types of Juvenile Arthritis	
Juvenile idiopathic arthritis (JIA)	Considered the most common form of arthritis, JIA includes six subtypes: oligoarthritis, polyarthritis, systemic, enthesitis-related, juvenile psoriatic arthritis or undifferentiated.
Juvenile dermatomyositis	An inflammatory disease, juvenile dermatomyositis causes muscle weakness and a skin rash on the eyelids and knuckles.
Juvenile lupus.	Lupus is an autoimmune disease. The most common form is systemic lupus erythematosus, or SLE. Lupus can affect the joints, skin, kidneys, blood and other areas of the body.
Juvenile scleroderma	Scleroderma, which literally means “hard skin,” describes a group of conditions that causes the skin to tighten and harden.
Kawasaki disease	This disease causes blood-vessel inflammation that can lead to heart complications.
Mixed connective tissue disease	This disease may include features of arthritis, lupus dermatomyositis and scleroderma, and is associated with very high levels of a particular antinuclear antibody called anti-RNP.
Fibromyalgia	This chronic pain syndrome is an arthritis-related condition, which can cause stiffness and aching, along with fatigue, disrupted sleep and other symptoms. More common in girls, fibromyalgia is seldom diagnosed before puberty.

Back

More common in girls than boys

Most commonly seen in adolescents. Estimated prevalence;

- 4% in 9-year-olds,
- 22% in 13-year-olds, and
- 36% in 15-year-olds.

Important to complete a good pain assessment, musculoskeletal, and neurological exam to diagnose:

- Tumor of the spine,
- Infection of the spine,
- Spondylosis
  - Mechanical back pain localized lateral to the midline, discomfort worsens with extension and rotation of the spine while standing
  - Repeated sub-clinical injury (cheerleading, gymnastics, football, diving, etc.)
  - Tenderness to palpation across the involved lever

Spondyloisthesis

Mechanical low back pain often presenting with abnormal gait and tight hamstrings

Herniated nucleus pulposis

- Low back pain with reticular radiation emanating from back and extending down to the ankle or foot
- Worse with coughing or sneezing
- May shift to get comfortable
- Passive straight leg maneuver will elicit nerve root irritation
- Diagnosis requires MRI

Musculo-ligamentous back pain

- Insidious onset of mechanical pain with poor, vague, or variable localization and poorly localized broad tenderness on PE
- Often associated with trigger points and tight musculature
- Requires exclusion of other disorders

Chronic Pain: Assessment

Significance

At risk groups for under recognition and inadequate treatment of chronic pain include:

- Infants and children
- Certain racial and ethnic groups
- Females
- Patients with a current or past history of substance abuse
- Cognitively impaired, nonverbal patients

Children and adolescents with chronic pain report:

- Depression, anxiety, hopelessness
- Impaired mobility, impaired physical function
- Weight gain (overweight and obesity) and weight loss
- Fatigue
- Decreased socialization, missed school and activities
- Family Dysfunction, parents missed work
- Cognitive dysfunction
- Sleep disturbances
- Increased health care utilization

Chronic Pain Assessment

Pain Associated Disability Syndrome (PADS)

- Pain is not just a number on an intensity scale, but how the pain impacts the child and family.
- Multidimensional assessment tools are used to assess chronic pain and include items about pain intensity, location, quality and the impact of pain on function and mood.
- The PADS model (1998) addresses many areas that can put a child at risk for disability with chronic pain.
- It is important to monitor risk factors and have safety nets in place for children to achieve optimal health behaviors.

After Bursch et al 1998

Chronic Pain: Treatment

Shelby

1. Hyperlordosis: The spine has natural forward and back curves. There's normally a thoracic kyphosis (apex posterior; hyperkyphosis is "hunchback") and a lumbar lordosis (apex anterior; hyperlordosis is "swayback")

2. Straight Left Raise Test (SLR): The SLR test is performed with the patient lying supine with legs fully extended. In the patient with lower back pain, evaluate the leg with pain. The examiner places one hand under the ankle of the affected leg and the other hand on the knee – and then lifts the ankle and flexes the thigh relative to the pelvis. The test is considered positive if pain is reproduced or increased in the lower back or leg. If the SLR is positive, further testing may be indicated.

Shelby is a 15-year-old obese female with back pain for 2 years without history of trauma.

- *Slight scoliosis <15 degrees*
- *Menarche at 12 years, regular, denies sexual contacts both voluntary and unwanted*
- *Previously healthy*
- *Mother with fibromyalgia, home on disability*
- *Sister is a freshman at a College 500 miles from home*

Pain now 8/10 pain.

- *Described as throbbing*
- *Pain not associated with menses*
- *Increased pain with activity*

The physical exam is unremarkable, other than it shows deconditioning common with chronic pain. The patient stands with most of her weight centered over the right foot, with the left hip and shoulder elevated and the left knee and hip slightly flexed. There is exaggerated lumbar lordosis (swayback).The abdominal muscles are weak, and there is marked tenderness to palpation over the lumbar region and lumbar paraspinal muscles. The patient has a negative Straight Leg Raise test.

Shelby

What are the treatment goals for Shelby's chronic pain?

The patient reports emotional distress related to pain and disability. She enjoyed school and socializing with friends but now really only has one friend who also has chronic abdominal pain from IBD. She is very worried about her future. She denies suicidal ideation or plan but does report some passive death wishes. Her sleep is poor: she has difficulty falling asleep, awakens in the night from pain and has difficulty awakening in the morning. She doesn't use alcohol or any medications except ibuprofen and acetaminophen, which don't work for her. She has been treated with Flexeril in the past and her mother has given her one of her hydrocodone/acetaminophen pills, but it didn't work either.

- *Depressed*
- *Passive death wishes, no active SI*
- *Friends "don't understand"*
- *Misses school or leaves school early due to pain at least 1 day/week*
- *Sleep disturbance*
- *No illicit drug use, has taken mother's pain medication*

Treatment Goals

Multimodal Treatment Plan

- Disease modification if appropriate
- Acknowledge feelings (e.g., grief, loss, frustration)
- Adopt a healthy lifestyle
- Set realistic individualized goals
- Active patient and family participation required
- Use a multimodal, interdisciplinary and self-management approach
  - Education, sleep, exercise, medications, nutrition, pain management skills, counseling
  - Flare plan

Although the typical goal of pain treatment is to relieve or significantly reduce pain intensity and severity, the primary goal for the patient and family suffering with chronic pain is to ***restore function***. Even when treatments do reduce pain intensity, they may not improve physical, emotional, and social function. Chronic pain treatment requires a coordinated, comprehensive, interdisciplinary approach which addresses all the dimensions of the child's condition and family functions. Active patient and family participation is needed to identify and attain individualized pain treatment goals.

Restore function

- Physical, emotional, social

Decrease pain

- Reduce or eliminate underlying cause of pain when possible
- Use individualized, multimodal, interdisciplinary, pain treatment plans

Correct secondary consequences of pain

- Postural deficits, weakness, overuse
- Maladaptive behavior, poor coping

Chronic Pain: Challenges

Chronic Pain: Challenges

Physical

- Multiple mechanisms/types of pain present
- Hurts to move initially

Psychosocial

- Pain alters school, social, family
- Depression, anxiety, anger

Behavioral

- Learned, reinforced pain behaviors
- Unconscious enabling by family, friends

Chronic pain treatment presents a complex challenge. Limiting physical activity to prevent or protect the child from pain leads to decreased physical activity, deconditioning, and secondary pains that leads the child and family to further limit physical activities.

It is counterintuitive to the child or family to consider the benefits of increasing physical activity as a pain treatment, and therefore it takes time to unlearn previously reinforced behaviors. Just as pain alters family and social interactions, these interactions will be difficult but necessary to change again to encourage more positive coping patterns.

Chronic Pain: Challenges

Care Providers:

- Multiple Providers
- Conflicting advice
- Disbelief from family & professionals
- Denial of problems
- Pain attributed to psychological factors

Patient and Family:

- Family & friends search and offer ideas about how to “cure” pain
- Difficulty adopting a rehabilitation model
- Adherence to recommendations
  - Costs of treatment
  - Time for treatment and transportation for therapies
- Intense fear of pushing too hard
- Difficulty tolerating the emotional distress
- Cultural beliefs
- Access to healthcare and treatment disparities

Providers may have been slow to refer to a specialist and reinforced an acute pain treatment model with repetitive diagnostic tests and multiple trials of medications and/or a wait and hope the patient grows out of it treatment plan. Families with lower resources and limited social supports have more difficulty accessing affordable care and a harder time adhering to care recommendations

Chronic pain Treatments

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Treatments include:

- Exercise and physical rehabilitation
- Cognitive behavioral therapy
- Lifestyle changes – sleep, diet, etc.
- School reintegration if needed
- Integrative medicine
- Pharmacotherapy
- Interventional treatments (infrequently used in children)
  - Trigger point injections
  - Nerve blocks

The results of studies of various treatments for chronic pain are sobering. “The best evidence for pain reduction averages roughly 30% in about half of treated patients, and these pain reductions do not always occur with concurrent improvement in function.”

Pharmacotherapy

Analgesics are not the primary treatment for pediatric chronic pain.

Opioids are rarely used to treat chronic pain in children and adolescents, unless there is an organic cause and strong evidence to support opioid therapy (for example repeated fractures in patients with osteogenesis imperfecta, or infections and disruptions in skin integrity with epidermolysis bullosa)

The CDC guidelines for chronic opioids use do not include recommendations for patients under 18 years of age due to lack of evidence.

Instead, medications may target depression, anxiety, and insomnia. Medications used may be off label for pediatric patients.

Pharmacotherapy Agreement

When medications are prescribed, a written agreement may help outline goals of therapy, expectations for treatment, including monitoring, follow-up, and other therapies.

"Medications Agreement"

- Clarifies the Partnership between the patient, family, and providers/team

Elements

- Goals of therapy
- Expectations for additional therapies
- Expectations for follow-up and monitoring
- Indications for tapering or discontinuing medications

Chronic Pain Treatments

How to manage pain flares

A chronic pain flare is a sudden increase in pain that may last from hours to days to weeks. It is distinct from "breakthrough pain."

Management should focus on identifying and addressing triggers:

- stress
- injury
- lack of sleep
- exercise
- hormones
- weather changes.

Interventional Treatments

Interventional treatments are rarely used in pediatric patients.

- Anesthetics/steroids may be used to block nerves or nerve roots/suppress inflammation.
- Not a cure, but may provide break in cycle of pain and temporary relief.

Biobehavioral Interventions

Biobehavioral interventions are the mainstay for chronic pain management for children and adolescents. These address the affective, cognitive, behavioral, sociocultural and spiritual dimensions of pain. They can diminish pain but focus on restoring function, supporting positive coping, and helping to relieve suffering that is associated with chronic pain.

Exercise improves physical functioning, decreases secondary sources of pain, and improves health.

Lifestyle changes are critical:

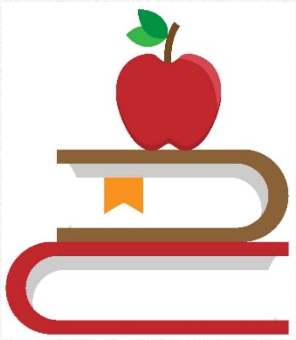
- Sleep hygiene
- Mental health
- Stress management

Promote ownership and sense of control:

- Yoga
- Tai Chi

Collaborative Self-management: Patients are responsible for creating and maintaining new life roles and coping with their emotional reactions to their conditions.

Education is Essential



* Author

Chronic pain is complex and frequently misunderstood.

Educate your patients about:

- the complexities of chronic pain
- the limitations of medical treatments

Empower your patients and their families with knowledge to actively participate in managing their pain and pain care rather than being victims of pain.

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Shelby

What would you include in her multimodal treatment plan given these goals?

Shelby's Treatment Goals

- *Goal: Increase Functional Activity*
- *Goal: Reduce Pain by 25% or more*
- *Goal: Diminish Psychological/Social Disruption*
- *Goal: Return to School*

In
Summary…

Key Points

- Perform a comprehensive initial pain assessment.
- Clarify difference between acute and chronic pain.
- Establish individualized multidimensional realized pain treatment goals with patient and family.
- Provide individualized, multimodal, interdisciplinary treatment plan.
- Educate patient and family about risks, benefits, limitations, and shared responsibilities for pain management plan.

Appendix

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